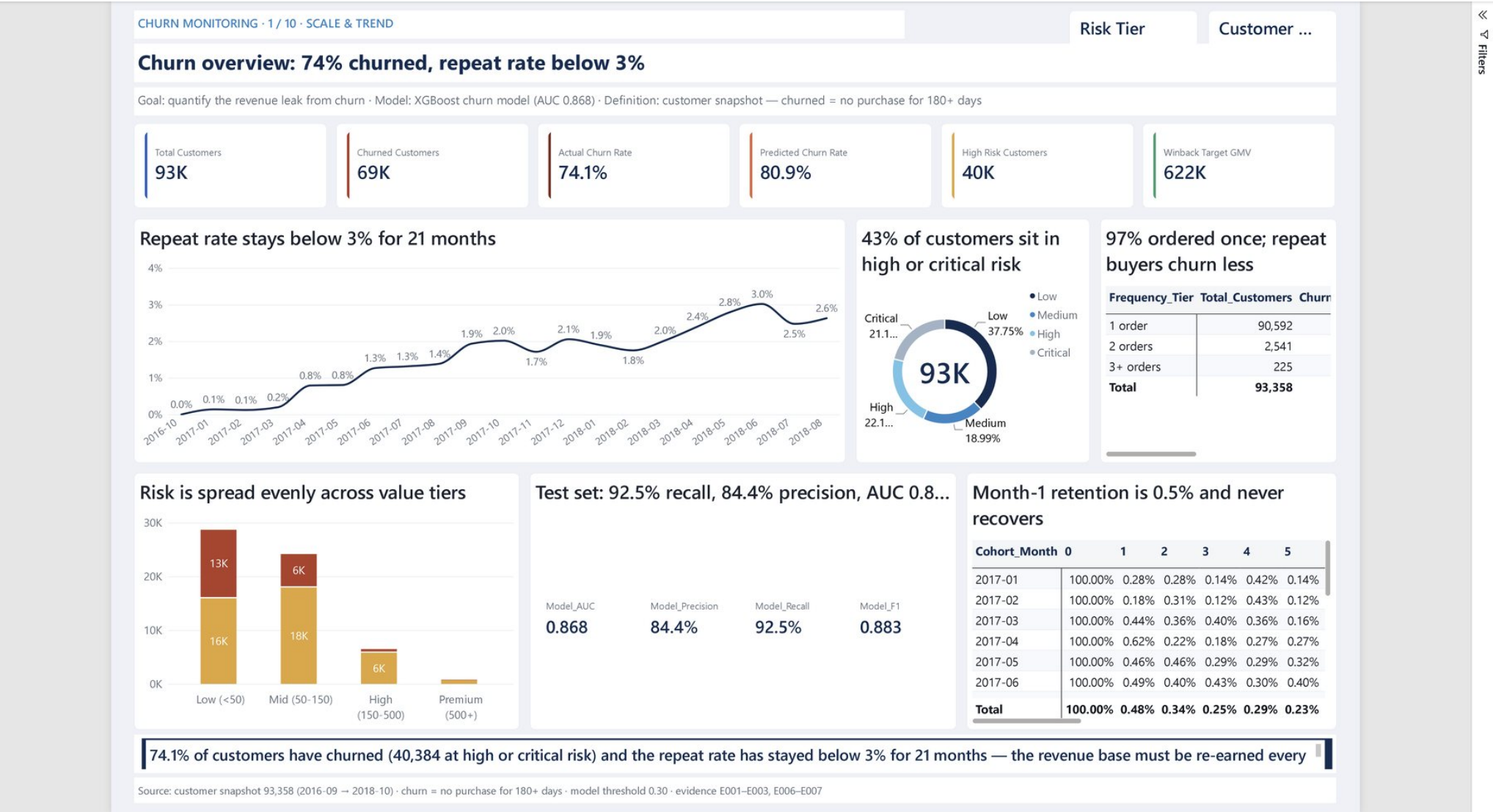


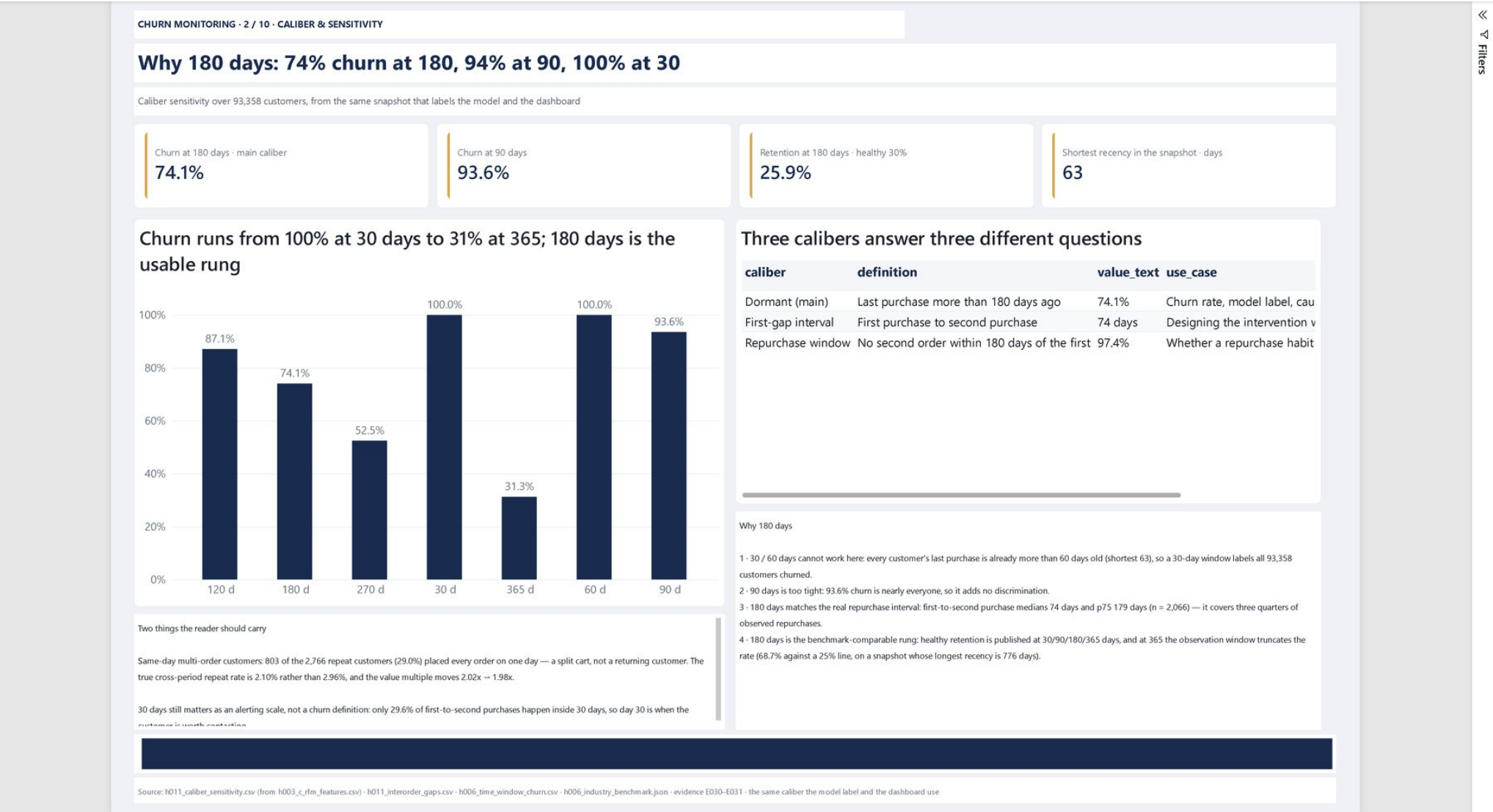
01 · Churn overview

How big is the leak, and is the trend improving?



02 · Churn caliber & sensitivity

What exactly is 'churn', and why 180 days?



03 · Who stays, who leaves

Which customers leave?



04 · Why they churn

What drives churn — global SHAP and per-customer TOP-1 driver.



05 · Cost of delay

What does a late delivery actually cost?

CHURN MONITORING · 5 / 10 · COST OF DELAY

Satisfaction breaks at 3-7 days late: low scores 26% to 62%

Five delay buckets over 99,992 orders, plus the slowest categories and states

low-score share, 3-7 days late

61.6%

low-score share, 7+ days late

78.3%

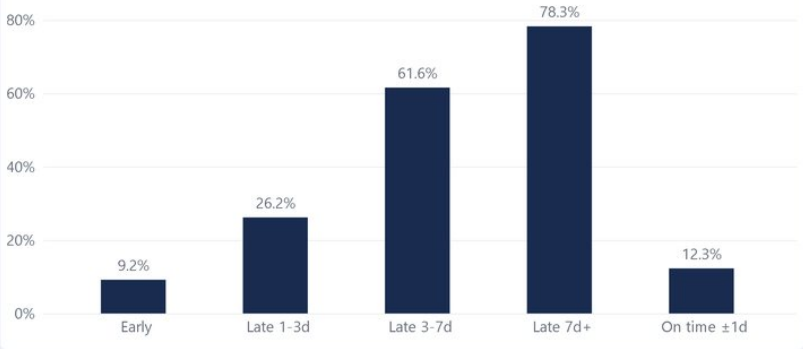
slowest state, average days

26.3

slowest category, average days

20.8

Low scores step from 26% to 62% between 3 and 7 days late



91.9% of orders arrive early; the late tail is where scores die

bucket_label	order_count	share	avg_review	low_pct
Early	88,552	91.90	4.294	9.2
Late 1-3d	1,335	1.39	3.493	26.2
Late 3-7d	1,790	1.86	2.305	61.6
Late 7d+	3,245	3.37	1.731	78.3
On time ±1d	1,431	1.49	4.035	12.3

AM is slowest at 26.3 days against a 12-day platform norm

group_name	n_orders	avg_days	avg_review	low_pct
AL	398	24.4	3.84	21.4
AM	145	26.3	4.22	13.8
BA	3,246	19.2	3.93	17.0
CE	1,276	21.2	3.94	17.3
DF	2,089	13.0	4.13	13.2
ES	1,978	15.6	4.08	13.8
GO	1,963	15.5	4.10	13.2
MA	716	21.4	3.84	19.8
MG	11,354	12.0	4.19	11.7
MS	710	15.6	4.17	13.1
MT	882	17.9	4.15	12.7

Office furniture is slowest at 20.8 days and 25.4% low scores

group_name	n_orders	avg_days	avg_review	low_pct
	1,533	12.7	3.94	19.4
audio	359	13.3	3.83	21.7
bebes	2,967	12.4	4.08	14.9
cama_mesa_banho	10,985	12.7	3.92	18.3
casa_conforto	430	13.5	3.85	19.8
cozinha	503	13.3	3.86	17.4

Source: h003_b4_delay_bucket.csv · h003_b1_delivery_status.csv · h003_b2_category_timing.csv · h003_b3_state_delivery.csv · evidence E037 · caliber note: on-time holds only 155 orders against 88,552 early ones

Filters

06 · Review themes (NLP)

What are customers complaining about?

CHURN MONITORING · 6 / 10 · VOICE OF THE CUSTOMER

Pain Point

Dissatisfact...

Voice of the customer: 60% of complaints are logistics

Source: low-score reviews (rating ≤ 2); pain point, sub-issue and improvement suggestion extracted by an LLM

Negative Reviews
300

Severe Dissatisfaction Reviews
12

Severe Dissatisfaction Share
4.0%

Logistics Pain Reviews
179

Avg Review Rating
1.2

Not received and partial delivery lead complain...

Pain Point	Count
Not received	80
Product mismatch	60
Partial delivery	55
Delivery delay	44
Product damaged	29
Other / unspecified	17
Refund issue	8
Seller communication	7

Severe dissatisfaction concentrates on non-delivery

Pain Point	Count
Delivery delay	28
Not received	64
Other / unspecified	10
Partial delivery	41
Product damaged	20
Product mismatch	48
Refund issue	9
Seller communication	7

Logistics accounts for 60% of complaints

Pain Point	Count
Delivery delay	44
Not received	80
Other / unspecified	17
Partial delivery	55
Product damaged	29
Product mismatch	60
Refund issue	8
Seller communication	7

Every complaint carries an actionable fix

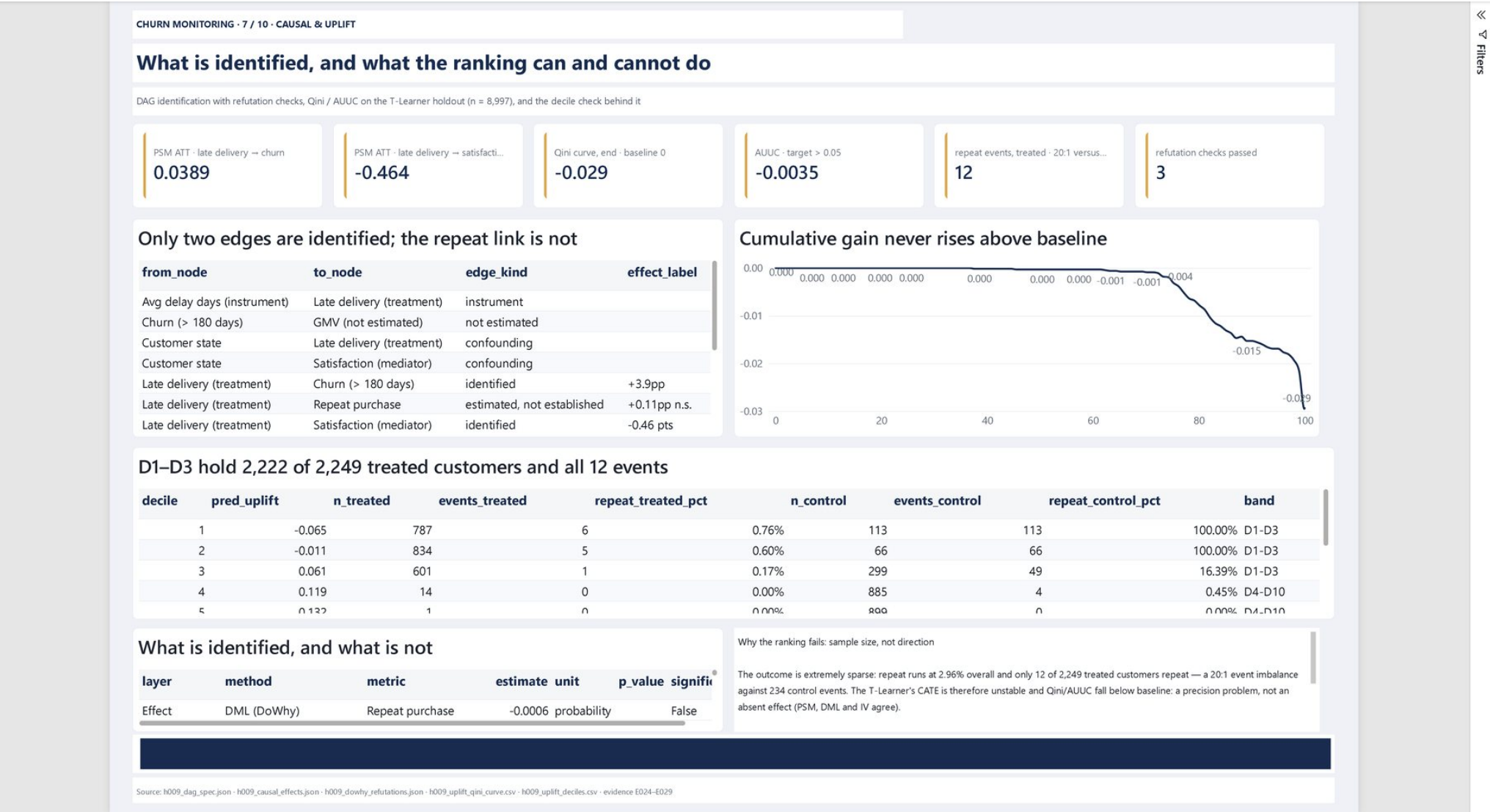
Pain_Point	Sub_Issue	Dissatisfaction_Level	Review_Keyword	Review_Translation
Delivery delay	approved_then_cancelled	Mild	POR DEMORA CANCELAMENTO	Order approved 10/06, cancelled 10/17 for delay, but then they ship it. Cant get contact via star
Delivery delay	at_post_office_wants_refund	Mild	produto esta nos correios... quero meu estorno	Paid shipping, product stuck at post office far from home. Wants refund.
Delivery delay	business_days_overdue	Mild	mais de 20 dias úteis	Ordered over 20 business days ago, still not delivered.
Delivery delay	chase_required	Dissatisfied	Tive que entrar em contato com a empresa pra receber	Had to chase the company to receive product. Wont buy this way again.
Delivery delay	correios_slow	Mild	dar 1 mês e o Correios não entregou	Regrets buying due to using Correios; almost 1 month and not delivered.
Delivery delay	extreme_delay	Mild	esperar mais de 60 dias	Never imagined waiting 60+ days; paying for a product that may never arrive is the same as not
Delivery delay	extreme_delay_caps	Mild	DEMOROU MUIIIITOOOOOOOO	Took a REALLY REALLY long time to arrive (caps for emphasis).
Delivery delay	first_issue_loyal	Mild	primeira vez que estou insatisfeito	Long-time lannister customer, first issue with delivery time.

60% of complaints are logistics (not received, partial, delay) — the same ranking the model produced, and only 4% are severe, so most are fixable service failures.

Source: 300 low-score reviews (rating ≤ 2), LLM extraction · evidence E015

07 · Causal & uplift

How much would an intervention move the needle, and for whom?



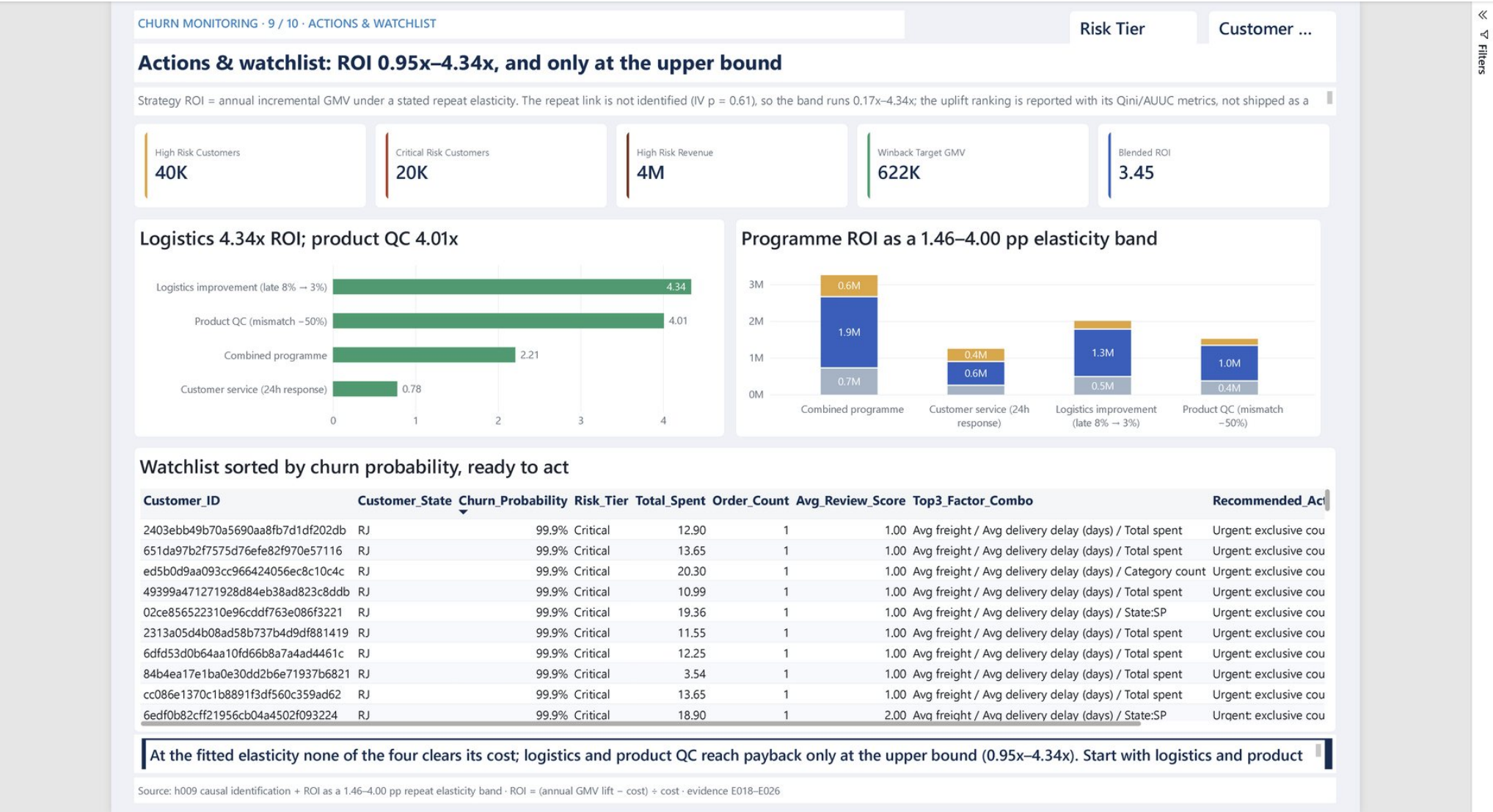
08 · Benchmark, structure & validation

How do we compare, and do the four estimators agree?



09 · Actions & watchlist

What to do, and who first?



10 · Appendix · order funnel & structure

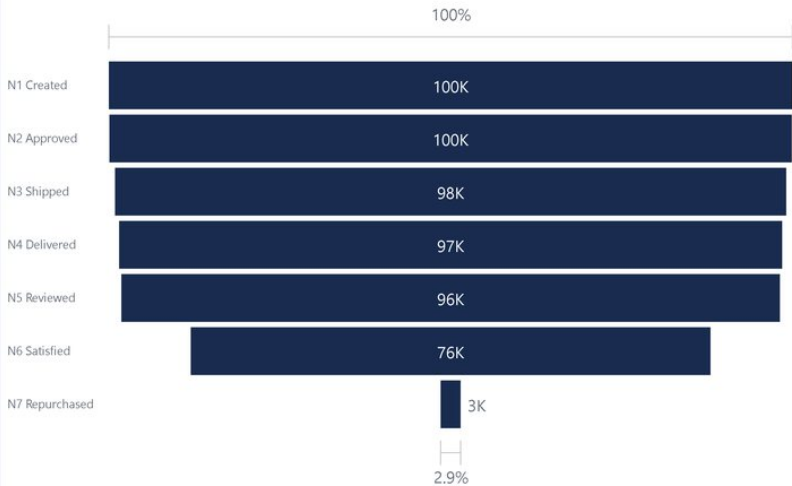
The full funnel and the monthly customer mix.

CHURN MONITORING · 10 / 10 · APPENDIX · ORDER FUNNEL & CUSTOMER STRUCTURE

Seven funnel nodes show where volume survives and where it does not

Snapshot 2016-09 → 2018-10 · 99,992 orders · step and cumulative conversion per node

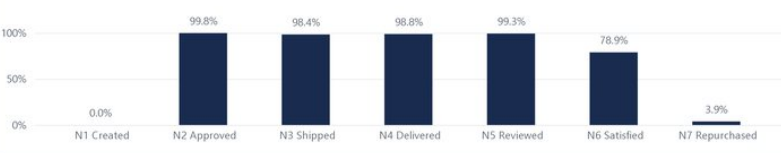
Satisfaction and repurchase are the only two real drops



Volume survives to review; satisfaction costs 21%, repurchase 96%

node_label	stage_count	step_conversion_pct	cumulative_conversion_pct
N1 Created	99,992	0.0%	0.0%
N2 Approved	99,831	99.8%	99.8%
N3 Shipped	98,185	98.4%	98.2%
N4 Delivered	96,990	98.8%	97.0%
N5 Reviewed	96,338	99.3%	96.4%
N6 Satisfied	76,033	78.9%	76.0%
N7 Repurchased	2,930	3.9%	2.9%

Only satisfaction and repurchase are real drops



Returning customers split three ways: consecutive buying is only 0.46% of buyers



Source: H004 funnel · snapshot 99,992 orders · evidence E004-E005 · FunnelNode measures rebuilt as progressive predicates

How this was produced. Human-led method, AI-assisted execution. The business problem, the analysis framework, the choice of method and the validation of every result are mine; AI handled implementation and drafting. Each figure is reconciled by an automated gate before it ships, and no claim reaches a client document without a human having checked it against its source snapshot.